

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}, \quad h^{\mu\nu} = g^{\mu\alpha} g^{\nu\beta} h_{\alpha\beta}$$

→ Can use $\eta^{\mu\nu}$ instead of $g^{\mu\nu}$ since this will come w/ higher order terms.

$$h_{\mu'\nu'} = \Lambda_{\mu'}^{\mu} \Lambda_{\nu'}^{\nu} h_{\mu\nu}$$

- Solve Einstein's Eq's in this spacetime:

$$R = 2\partial_{\mu}\partial_{\nu}h^{\mu\nu} - \square h$$